

Grid Network: Feature Extraction in Anisotropic Perspective for Hyperspectral Image Classification

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Abstract—Abundant spectral signatures and spatial characteristics embedded in hyperspectral (HS) images enable the fine identification of land covers, attracting plenty of studies on feature extraction (FE) and feature utilization. Nevertheless, the high representative spectral and spatial features in the HS cube are unevenly distributed, which is failed to consider by many current methods. To conquer this shortcoming, we rethink the FE of HS images from an anisotropic perspective and propose a novel model called grid network (GNet) for HS image classification (HSIC). Beyond representing spectral-spatial features in three classic paradigms (simultaneously, hierarchically, and separately), GNet is capable of learning them in two new processes: multistage and multipath. In this way, spectral and spatial features can be fully and balanced explored. More significantly, to make full use of low- and high-level features and avoid the existing semantic gap, we devise a spectral-spatial cross-level feature fusion (S^2CLF^2) module to model the relationship between them. Extensive experiments, implemented on three HS datasets, demonstrate that the proposed GNet enables to acquire promising classification performance compared with state-of-the-art methods. The codes of this work will be available at https://github.com/zhonghaochen/GNet_Master for the sake of reproducibility.

Index Terms—Anisotropic, feature fusion, hyperspectral (HS) images, semantic gap, spectral-spatial feature.

I. INTRODUCTION

HYPERSPECTRAL (HS) image has been broadly investigated in Earth observation (EO) due to its excellent spectral signature description ability besides obtaining the spatial distribution of land covers [1]. Specifically, the hundreds of narrowbands' spectral information crossing the entire electromagnetic spectrum range included in HS images allow for the fine-grained identification of materials [2]. Accordingly, pixelwise land cover classification equipped with HS image has been a vibrant research topic. Despite the great potential of rich spectral features, high-dimensional information extraction

and how to balance the extraction of spatial and spectral features can be highly challenging.

In the early phase of HS image classification (HSIC), plenty of advanced methods using subspace learning and statistical theory have been proposed. The performance of these traditional methods has been demonstrated to be acceptable, especially in the case of limited samples [3]. However, the limited feature extraction (FE) ability of these methods prevents the acquisition of superior discriminative features and hence hinders classification performance. In recent years, deep learning (DL) has been widely explored in HSIC due to its powerful ability of extracting intrinsic and potentially features. Specifically, several DL-based backbone networks have been thoroughly and successfully investigated in HSIC, such as convolutional neural networks (CNNs) [4], generative adversarial networks (GANs) [5], graph neural networks (GNNs) [6], [7], and transformers [8]. Among these backbone networks, CNN is one of the most popular tool in the HSIC tasks. In general, depending on the order in which features (spectral or spatial) are extracted from HS images, the current mainstream CNN-based classification methods can be divided into the following three categories: 1) extracting simultaneously; 2) extracting hierarchically; and 3) extracting separately. Fig. 1 provides a brief illustration of these three architectures.

For the spectral-spatial feature simultaneously extracting, a series of 3-D CNN-based methods are developed. Li et al. [4] cascaded two 3-D convolutional kernels with a fully connected (FC) layer to construct an effective spectral-spatial FE model, yielding promising results. Ahmad et al. [9] performed incremental principal component analysis (iPCA) on the original HS images and fed them in a shallow 3-D CNN to reduce computational complexity and obtained higher accuracy. However, purely using 3-D CNN inevitably resulted in numerous learnable parameters, which motivated Roy et al. [10] to hierarchically adopt 3-D and 2-D CNN to design HybridSN for excavating spectral and spatial features hierarchically with fewer parameters. However, this progressive FE idea ignores the potential correlation information between spectral and spatial features, and the original distribution of later extract features might also be compromised. The third FE strategy is commonly implemented by deliberately developing different subnetworks for spectral and spatial features [11]. For instance, Xue et al. [12] extended the original residual unit to a hierarchical residual unit to realize multiscale spectral and spatial FE, respectively, yielding excellent performance.

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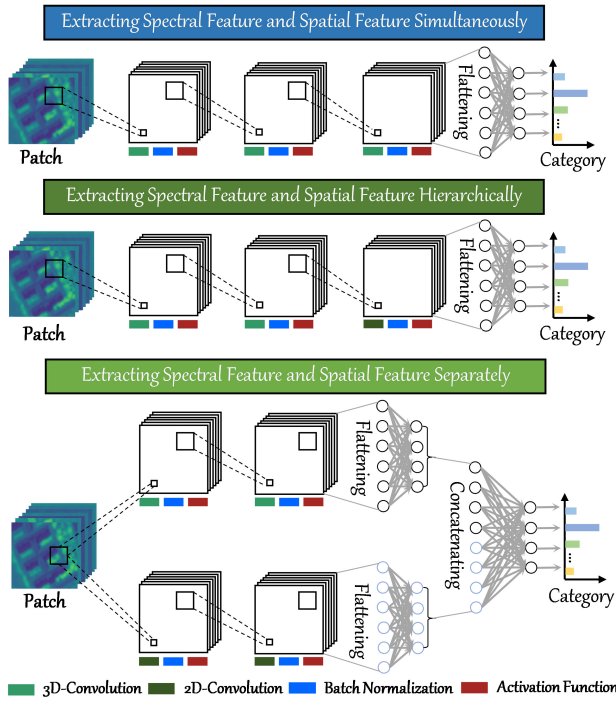


Fig. 1. Overview of currently three CNN-based spectral and spatial FE paradigms for the HSIC task.

More recently, Song et al. [8] combined CNN and transformer to construct two dedicated networks for spectral and spatial feature representation, separately.

Although promising classification performance has been obtained by the aforementioned three paradigms and their variants, their ability in balancing the extraction of spectral and spatial features remains inadequate. Specifically, HS images acquired by different sensors in multifarious scenes might possess various qualitatively inhomogeneous spatial and spectral features (the HS cube can be understood as anisotropic data [13]), which makes them different in terms of extraction difficulty. Consequently, the problem of imbalanced representation of spectral interband and spatial context information still exists inside the aforementioned methods (serial or parallel combination), limiting the classification performance.

Being aimed at this limitation, in this letter, we rethink the CNN-based classification process from an anisotropic data perspective and propose a grid network (GNet) for the HSIC tasks. Unlike the aforementioned methods, GNet aims to represent spectral and spatial features in a more balanced manner. In addition, a spectral–spatial cross-level feature fusion (S^2CLF^2) module is designed in GNet to merge low-level texture and high-level semantic features. More specifically, our major contributions are the following.

- 1) We study HS image FE from the anisotropic perspective for the first time and particularly propose GNet for HSIC. Specifically, the spectral and spatial features can be explored in multiple processes, which can empower the model with more freedom to extract balanced representation for classification.
- 2) We devise a simple and generic module in GNet, called S^2CLF^2 , to fuse low-level and high-level features under

the guidance of interrelated information, alleviating the affection of semantic gap.

- 3) We thoroughly evaluate the performance of the proposed GNet on three benchmarks with standard training and test samples, acquiring outstanding results.

II. METHODOLOGY

In this section, the detail of our proposed GNet will be well-described, first. Then, S^2CLF^2 is proposed to make GNet more applicable to the HSIC task.

A. GNet

We aim at developing a flexible and end-to-end CNN-based method with a focus on balancing the spectral and spatial feature excavation, making it capable to provide discriminative features for HSIC. To this end, a network is devised by not only flexibly combining the three aforementioned spectral–spatial FE strategies but also introducing two novel feature explore processes (multistage and mutipath). Therefore, the spectral and spatial features embedded in the HS images can be adaptively collected by learning for the end classification. Fig. 2(a) illustrates the architecture of the proposed GNet in the classification of HS images.

For the sake of comprehensiveness exploring abundant spectral signatures and its location-dependent spatial context, patchwise training strategy is adopted here. Specifically, principal component analysis (PCA) is first adopted to transform the original HS image $X \in \mathbb{R}^{H \times W \times B}$ into $X' \in \mathbb{R}^{H \times W \times P}$ by retaining the main features in spectral dimension, where H , W , B , and P represent the spatial height, width, number of raw spectral bands, and retained principal components of spectral information, respectively. Then, X' is cropped into HW cubes in spatial dimension as the input of GNet. Therefore, each cube is formed by each pixel in X' and its $S \times S$ neighborhood pixels, which can be further represented as $x_i \in \mathbb{R}^{S \times S \times P}$, $i \in (1, 2, \dots, HW)$. As depicted in Fig. 2(a), the proposed GNet is a 3×3 grid-like model in which every grid is composed of two 3-D convolutional kernels of size $3 \times 3 \times 1$ [the green cube in Fig. 2(a)] and two 3-D convolutional kernels of size $1 \times 1 \times 3$ [the blue cube in Fig. 2(a)] in four edges respectively. As a result, the spectral and spatial features are extracted twice as the feature maps pass through a mesh. By connecting the output of each grid through matrix addition operations, the grid architecture of GNet is naturally constructed. It is worth noting that the order of spectral and spatial FE varies in different grids, resulting in the extraction of more discriminative features. Moreover, the GNet composed of these different grids brings us various subsets, which not only include the aforementioned three spectra–spatial representation processes but also two novel processes [see Fig. 2(b)]. The first new process, multilevel FE, is motivated by the joint spectral–spatial latent features that can be further excavated after the spectral and spatial features have been explored hierarchically. For another process, multipath FE can better facilitate the model to perform FE at different depths, thus generating multilevel features. On the whole, benefiting from these diverse processes, the selection of spectral and spatial

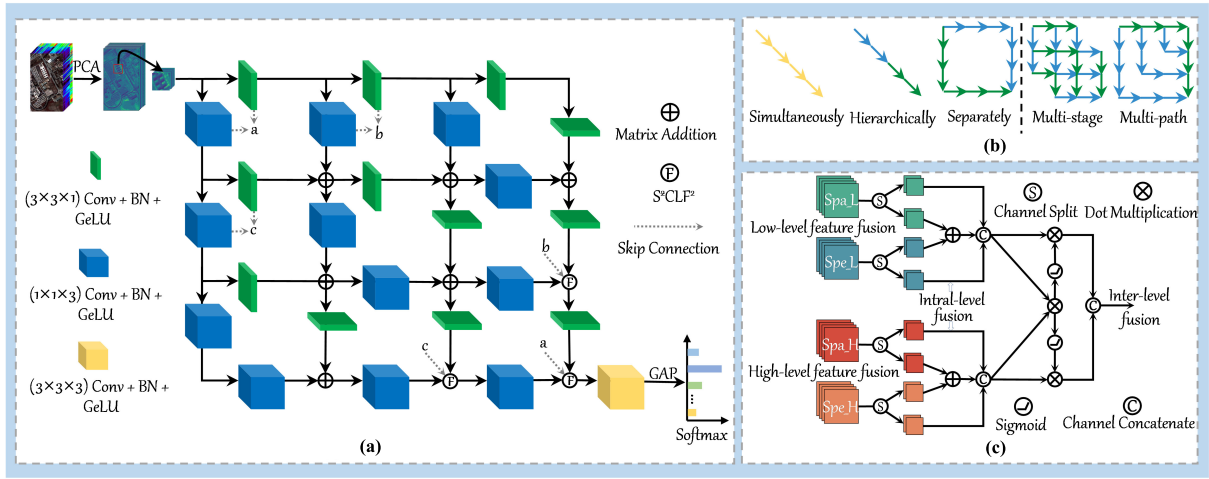


Fig. 2. Overview of the proposed GNet for the HSIC task. GNet consists of a multiple spectral-spatial FE process and S²CLF² module, making it better applicable to FE of HS images from anisotropic perspective. (a) GNet. (b) Multiple process of feature maps. (c) Spectral-spatial cross-level feature fusion.

representation process is unconstrained, which can adaptively balance the spectral and spatial FE resulting in a stronger model for anisotropic HS images.

Moreover, because the above decoupling extraction of spectral and spatial features may be weak in spectral-spatial joint feature exploration, we have added an extra 3-D convolution option of size $3 \times 3 \times 3$ at the end. The obtained discriminative spectral-spatial feature maps $\mathbf{M}$ are then flattened into the feature vector using a global average pooling (GAP) operation. Finally, an FC layer with SoftMax activation function is adopted to acquire the final decision vector $\mathbf{V}_d$, formulated as follows:

$$\mathbf{V}_d = \delta(\phi(\varphi(\mathbf{M}))) \quad (1)$$

where the operators $\delta(\cdot)$, $\phi(\cdot)$, and $\varphi(\cdot)$, respectively, represent the SoftMax function, FC layer, and GAP layer.

B. Spectral-Spatial Cross-Level Feature Fusion

Fusing low-level features in shallow with high-level features in deep has been demonstrated to be a powerful way in deep networks, which can enrich the feature representation and improve feature utilization. However, the great semantic gap among low- and high-level features would undermine classification results if they are fused in a naive way. This is a significant challenge for effectively leveraging them for boosting the classification performance. To this end, S²CLF², a novel fusion strategy, is devised in GNet to fuse features more properly with the high semantic gap [detailed in Fig. 2(c)].

As detailed in Section II-A, the spectral and spatial features of the HS image are explored decoupled and hierarchically, thus letting $\mathbf{Spe}_L$ and $\mathbf{Spa}_L$ be the outputs of low-level features while $\mathbf{Spe}_H$ and $\mathbf{Spa}_H$ denote high-level features. Subsequently, S²CLF² will first obtain low- and high-level joint spectral-spatial features by intralevel feature fusion, and then fuse low- and high-level features by feature embedded. Specifically, in the intralevel feature fusion stage (taking low-level feature fusion as an example), the low-level spectral

and spatial features are first equally split into two along the channel. These can be expressed as

$$(\mathbf{Spe}_L[0], \mathbf{Spe}_L[1]) = \varsigma(\mathbf{Spe}_L) \quad (2)$$

$$(\mathbf{Spa}_L[0], \mathbf{Spa}_L[1]) = \varsigma(\mathbf{Spa}_L) \quad (3)$$

where $\varsigma(\cdot)$ denotes the channel split operation. Then, the low intralevel fusion feature $\mathbf{SpeSpa}_L$ can be obtained as

$$\mathbf{SpeSpa}_L = [\mathbf{Spe}_L[0], (\mathbf{Spe}_L[1] + \mathbf{Spa}_L[1]), \mathbf{Spa}_L[0]] \quad (4)$$

where $[\cdot]$ denotes channel concatenation. Similarly, the high-intralevel fusion feature can be acquired and formulated as $\mathbf{SpeSpa}_H$. According to (4), it is worth noting that the obtained intralevel fusion feature not only contains the joint spectral-spatial features but also preserves the spectral and spatial features individually, which will enable subsequent interlevel integration to yield richer features. After intralevel feature fusion, the interlevel feature relation embed strategy is adopted to fuse low- and high-level features and alleviate the semantic gap. First, the pixelwise relation map of low- and high-level features is calculated by performing matrix dot multiplication to alleviate the semantic gap in further fusing steps. Naturally, the relation weight matrix $\mathbf{F}_r$ can be given by

$$\mathbf{F}_r = \delta(\mathbf{SpeSpa}_L \times \mathbf{SpeSpa}_H) \quad (5)$$

where $\xi(\cdot)$ denotes the Sigmoid activation function. Then, we multiply it back to low- and high-level features, respectively, to highlight their significant common features. Consequently, the commonality of low- and high-level characteristics will be more pronounced and the semantic gap would be suppressed. Formally, the enhanced low-level features $\mathbf{F}_L$ and high-level features $\mathbf{F}_H$ are expressed, respectively, as

$$\mathbf{F}_L = \mathbf{SpeSpa}_L \times \mathbf{F}_r \quad (6)$$

$$\mathbf{F}_H = \mathbf{SpeSpa}_H \times \mathbf{F}_r. \quad (7)$$

Finally, the high representation spectral-spatial cross-level fused features are obtained by concatenating $\mathbf{F}_L$ and $\mathbf{F}_H$ along the channel dimension, formulated as follows:

$$\mathbf{F} = [\mathbf{F}_L, \mathbf{F}_H]. \quad (8)$$

In our proposed GNet, three S^2CLF^2 s are adopted to fuse three groups of low- and high-level features (denoted as a–c) to collect abundant cross-level fused features, detailed in Fig. 2.

III. EXPERIMENTS

A. Datasets and Experimental Setup

Three well-known benchmark datasets are adopted to evaluate the performance of the proposed GNet, which are Indian Pines (IP), University of Pavia (PU), and Houston2013¹² (HS). Remarkably, standard training and test samples referencing to more recent studies [2], [6] are used for parameter training to properly demonstrate the performance.

To fully evaluate the performance of the proposed GNet, eight representative algorithms are selected for comparison, including support vector machine (SVM), random forest (RF), 3-D CNN [4], multiscale residual network (MSRN) [14], HybridSN [9], lightweight spectral–spatial attention network (LSSAN) [15], HResNetAM [12], and BS2T [8]. All the experiments are implemented on Tensorflow-gpu 2.5.0 platform using a workstation with RTX 3090 GPU. For our method, the Adam optimizer is adopted to optimize parameters, and the minibatch sizes are set to 32, 32, and 128, respectively, for three datasets. The initial learning rate is set to 0.001 and updated using the “Inverselime” learning rate policy, in which decay steps and the decay rate are set to 10 and 0.01, respectively. In the training process, the maximum training epochs are set to 150. For all the datasets, furthermore, the input size of GNet is set to $11 \times 11 \times 10$. Fairly, for other comparison methods, we completely refer to the original work to build the model and adjust the hyperparameters to achieve the optimum in this experimental platform. Moreover, the classification performance is evaluated quantitatively by three commonly applied assessment criteria: overall accuracy (OA), average accuracy (AA), and kappa coefficient.

B. Ablation Experiment

We thoroughly investigate the classification performance gain of our proposed S^2CLF^2 in terms of OAs using various interlevel fusion strategies in the GNet. Therefore, we consider the effect of nonfusion of low- and high-level features and four fusion strategies (i.e., addition (A), concatenation (C), elemental multiplication (M), and our S^2CLF^2) on the experimental results. Detailed classification results are reported in Table I. As can be observed, not all the fusion strategies can effectively fuse shallow and deep features to obtain more comprehensive feature representation and achieve better performance. In other words, due to the semantic gap between low- and high-level features, inappropriate fusion methods may also achieve lower classification results than nonfusion (e.g., fusion strategy M on the PU dataset). Notably, by plugging our proposed cross-level fusion strategy into GNet, a stable performance improvement is yielded on the three datasets. This suggests that the obtained correlation weight matrix can indeed mitigate the semantic

TABLE I
CLASSIFICATION PERFORMANCE (OA) OF THE GNet WITHOUT OR WITH
DIFFERENT INTERLEVEL FUSION STRATEGIES

Datasets	W/O	A	C	M	Proposed
IP	94.35%	94.55%	94.52%	94.45%	95.07%
PU	92.55%	93.81%	93.24%	92.34%	94.31%
HS	89.19%	88.91%	89.77%	88.96%	90.31%

gap between low- and high-level features and facilitate further fusion.

C. Classification Performance Analysis

Table II quantitatively reports the classification performance of different methods in terms of OA, AA, and Kappa coefficient for the IP, PU, and HS datasets, respectively.

Overall, the classification scores on three datasets yielded by RF and SVM are similar and relatively poor due to only spectral signatures being used. On the contrary, by virtue of the utilization of spatial information and the powerful feature representation ability of DL, other DL-based models obtain more competitive performance compared with the traditional methods based on manual features. Unlike 3-D CNN, MSRN introduced a multiscale FE mechanism, yielding better classification performance. Not surprisingly, the performance of LSSAN generally outperforms that of HybridSN, because the additional attention strategy can facilitate useful features being intensified. Moreover, profited by the global information collected by transformer architectures, BS2T shows better performance than HResNetAM. However, it is worth mentioning that HResNetAM yields higher classification results on high spatial resolution datasets (e.g., PU dataset) due to the hierarchical multiscale learning strategy.

More remarkably, our proposed GNet achieves a more outstanding performance than other state-of-the-art DL-based methods, especially in terms of generalizability. Although the aforementioned methods based on the existing three FE paradigms are carefully designed, they lack the ability to adaptively extract spectral and spatial features in the FE process. To overcome this deficiency, the proposed GNet elegantly investigates spectral and spatial features in HS images in a more balanced way, dramatically boosting the classification scores. As expected, the GNet achieves optimal performance on all three datasets with different spectral and spatial resolutions. Moreover, we further investigate the total parameters of different methods (see Table II). As reported, although LSSAN possesses a more lightweight structure, limited parameters affect its ability to fit the data, especially on the PU dataset.

Since the labels of land covers are often difficult to obtain in the real-world scenario, we also explored the impact of the proportion of labeled samples on classification results. Specifically, different percentages of training samples from 10% to 100% per class are randomly selected from spatially disjoint samples for training all the models. As shown in Fig. 3, on the whole, the classification accuracy rises with the use of more training samples. Remarkably, it can be observed that our proposed GNet outperforms all the comparison methods, especially only 10% labeled samples used. Interestingly, a noteworthy trend emerges in Fig. 3. That is,

¹<http://www.grss-ieee.org/community/technical-committees/data-fusion/2013-ieee-grss-data-fusion-contest/>

²The cloud-free version was provided by Prof. N. Yokoya from the University of Tokyo and RIKEN AIP.

TABLE II
QUANTITATIVELY COMPARISON OF DIFFERENT STATE-OF-THE-ART CLASSIFICATION ALGORITHMS ON THREE DATASETS.
THE BOLD ENTITIES INDICATE THE BEST ONE

Methods		Indian Pines			University of Pavia			Houston2013			Total Parameters (PU Dataset)
		OA (%)	AA (%)	Kappa	OA (%)	AA (%)	Kappa	OA (%)	AA (%)	Kappa	
Classic	RF	69.71	76.53	0.6602	69.75	80.27	0.6198	77.52	80.39	0.7579	-
	SVM	72.38	83.21	0.6891	70.85	84.59	0.6455	76.91	78.99	0.7501	-
Simultaneously	3-D CNN	88.64	94.01	0.8701	88.84	93.96	0.8458	85.05	86.98	0.8378	58.5M
	MSRN	94.33	96.22	0.9374	91.38	88.01	0.8786	86.18	87.78	0.8502	1.44M
Hierarchically	HybridSN	91.01	95.62	0.8986	83.61	89.01	0.7876	81.92	83.58	0.8037	48.4M
	LSSAN	94.56	96.71	0.9382	84.59	90.52	0.7981	88.99	91.12	0.8879	0.45M
Separately	HResNetAM	86.24	92.83	0.8434	92.42	93.62	0.8992	86.43	87.64	0.8529	1.15M
	BS2T	90.47	91.33	0.8859	87.09	88.38	0.8587	87.69	84.34	0.8273	2.02M
Anisotropically	GNet	95.07	97.55	0.9437	94.32	93.29	0.9228	90.31	91.95	0.8948	0.84M

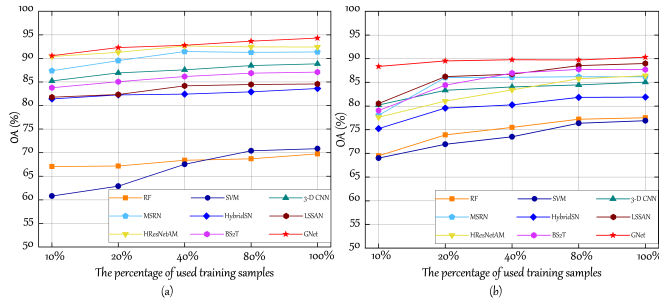


Fig. 3. Classification performance (OA) obtained by different methods with varying percentages of training samples on (a) PU dataset and (b) HS dataset.

there is no significant increase in classification accuracy for all the methods as the training sample size increases from 10% to 100%, unlike the results reported in other studies. Moreover, not all the methods achieve optimal classification performance when the training sample size is maximized, which is different from the conclusion obtained from intuition. These phenomena might be carefully explained by the high intraclass similarity of the training samples provided by the GRSS DASE website,³ which cannot provide more diversified features.

IV. CONCLUSION

Since the discriminative features in HS images are not uniformly distributed in spectral and spatial dimensions, it is vital to drive the model to explore these two heterogeneous features in a more flexible manner. In this letter, accordingly, a novel FE network based on anisotropic perspective is proposed for HSIC, called GNet. Specifically, the proposed GNet can explore spectral and spatial features in multiple processes and leverages them in a well-balanced manner. Furthermore, a plug-and-play S^2CLF^2 is constructed to degrade semantic gaps between low- and high-level features and then fuse them to boost classification performance. Our extensive experimental results, implemented on three widely used datasets, demonstrate that our proposed GNet is effective and superior.

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REFERENCES

- [1] Z. Chen et al., "Global to local: A hierarchical detection algorithm for hyperspectral image target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5544915.
- [2] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.
- [3] B. Rasti et al., "Feature extraction for hyperspectral imagery: The evolution from shallow to deep: Overview and toolbox," *IEEE Geosci. Remote Sens. Mag.*, vol. 8, no. 4, pp. 60–88, Dec. 2020.
- [4] Y. Li, H. Zhang, and Q. Shen, "Spectral-spatial classification of hyperspectral imagery with 3D convolutional neural network," *Remote Sens.*, vol. 9, no. 1, p. 67, Jan. 2017.
- [5] L. Zhu, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Generative adversarial networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5046–5063, Sep. 2018.
- [6] D. Hong, J. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.
- [7] Z. Chen, G. Wu, H. Gao, Y. Ding, D. Hong, and B. Zhang, "Local aggregation and global attention network for hyperspectral image classification with spectral-induced aligned superpixel segmentation," *Expert Syst. Appl.*, vol. 232, Dec. 2023, Art. no. 120828.
- [8] R. Song, Y. Feng, W. Cheng, Z. Mu, and X. Wang, "BS2T: Bottleneck spatial-spectral transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5532117.
- [9] M. Ahmad, A. M. Khan, M. Mazzara, S. Distefano, M. Ali, and M. S. Sarfraz, "A fast and compact 3-D CNN for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
- [10] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.
- [11] H. Gao, Z. Chen, and F. Xu, "Adaptive spectral-spatial feature fusion network for hyperspectral image classification using limited training samples," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 107, Mar. 2022, Art. no. 102687.
- [12] Z. Xue, X. Yu, B. Liu, X. Tan, and X. Wei, "HResNetAM: Hierarchical residual network with attention mechanism for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3566–3580, 2021.
- [13] Z. Dong et al., "MNet: Rethinking 2D/3D networks for anisotropic medical image segmentation," 2022, *arXiv:2205.04846*.
- [14] H. Gao, Y. Yang, C. Li, L. Gao, and B. Zhang, "Multiscale residual network with mixed depthwise convolution for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3396–3408, Apr. 2021.
- [15] Y. Cui, J. Xia, Z. Wang, S. Gao, and L. Wang, "Lightweight spectral-spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5510114.

³<http://dase.grss-ieee.org>